API SPECIFICATION

Complete REST API baseline for the MCI assistive application

AI-Powered Multimodal Assistive Application for People with Mild Cognitive Impairment (MCI)

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1. API Conventions

Item	Baseline
Style	REST/JSON over HTTPS on supported deployment
Base URL	/api/v1
Authentication	Bearer access token; refresh mechanism may be used
Content-Type	application/json unless multipart upload is explicitly required
IDs	UUID recommended
Time	ISO 8601 timestamps; patient timezone used for display/reminders
Errors	Consistent JSON error object
Pagination	page/page_size or cursor for list endpoints

2. Standard Response Formats

Success:

```
{"success": true, "data": {...}, "meta": {...}}
```

Error:

```
{"success": false, "error": {"code": "ERROR_CODE", "message": "Human-readable message"}}
```

3. Authentication APIs

Method	Endpoint	Auth	Purpose
POST	/auth/login	Public	Authenticate user.
POST	/auth/refresh	Refresh	Issue/refresh access token.
POST	/auth/logout	Bearer	Invalidate/logout session if supported.
GET	/auth/me	Bearer	Return current user and role.

4. Patient and Caregiver APIs

Method	Endpoint	Auth	Purpose
GET	/patients/me	Patient/Caregiver	Current patient profile or authorized patient context.
GET	/patients	Caregiver	List authorized patients.
GET	/patients/{patient_id}	Caregiver	Get patient profile.
PATCH	/patients/{patient_id}	Caregiver	Update patient profile.
GET	/caregivers/me	Caregiver	Current caregiver profile.
GET	/patients/{patient_id}/caregivers	Caregiver	List authorized caregivers.
POST	/patients/{patient_id}/caregivers	Caregiver/Admin	Authorize caregiver.
DELETE	/patients/{patient_id}/caregivers/{caregiver_id}	Caregiver/Admin	Revoke authorization.

5. Schedule and Calendar APIs

Method	Endpoint	Auth	Purpose
GET	/patients/{patient_id}/schedule	Patient/Caregiver	Get daily/recurring schedule.
POST	/patients/{patient_id}/schedule	Caregiver	Create routine item.
PATCH	/schedule/{schedule_id}	Caregiver	Update routine.
DELETE	/schedule/{schedule_id}	Caregiver	Deactivate/delete routine.
GET	/patients/{patient_id}/appointments	Patient/Caregiver	List appointments.
POST	/patients/{patient_id}/appointments	Caregiver	Create appointment.
GET	/appointments/{appointment_id}	Patient/Caregiver	Get appointment.
PATCH	/appointments/{appointment_id}	Caregiver	Update appointment.
DELETE	/appointments/{appointment_id}	Caregiver	Delete appointment.

6. Medication and Reminder APIs

Method	Endpoint	Auth	Purpose
GET	/patients/{patient_id}/medications	Patient/Caregiver	List medications.
POST	/patients/{patient_id}/medications	Caregiver	Create medication.
GET	/medications/{medication_id}	Patient/Caregiver	Get medication.
PATCH	/medications/{medication_id}	Caregiver	Update medication.
DELETE	/medications/{medication_id}	Caregiver	Deactivate medication.
POST	/medications/{medication_id}/schedules	Caregiver	Create medication schedule.
PATCH	/medication-schedules/{schedule_id}	Caregiver	Update schedule.
DELETE	/medication-schedules/{schedule_id}	Caregiver	Deactivate schedule.
GET	/patients/{patient_id}/reminders	Patient/Caregiver	List reminders.
POST	/reminders/{reminder_id}/acknowledge	Patient	Mark reminder acknowledged.

7. Familiar People and Face Recognition APIs

Method	Endpoint	Auth	Purpose
GET	/patients/{patient_id}/people	Patient/Caregiver	List familiar people.
POST	/patients/{patient_id}/people	Caregiver	Register familiar person.
GET	/people/{person_id}	Patient/Caregiver	Get person profile.
PATCH	/people/{person_id}	Caregiver	Update person.
DELETE	/people/{person_id}	Caregiver	Deactivate/revoke person.
POST	/people/{person_id}/face-embeddings	Caregiver	Register supported face representation.
POST	/face/recognize	Patient	Recognize a face from supported input.

8. Object Intelligence APIs

Method	Endpoint	Auth	Purpose
POST	/objects/detect	Patient	Submit supported image/frame for object detection.
GET	/patients/{patient_id}/objects	Patient/Caregiver	List tracked objects.
POST	/patients/{patient_id}/objects	Patient/Caregiver	Create/register object.
GET	/objects/{object_id}/observations	Patient/Caregiver	List object observations.
GET	/objects/{object_id}/last-seen	Patient/Caregiver	Return latest valid observation.
POST	/objects/query	Patient	Natural-language/object-name query for last-seen retrieval.

9. Conversation and RAG APIs

Method	Endpoint	Auth	Purpose
POST	/conversations	Patient	Create conversation session.
POST	/conversations/{conversation_id}/messages	Patient	Store a message/transcript segment.
POST	/memory/ingest	Patient/System	Create searchable memory from eligible content.
POST	/memory/search	Patient/System	Semantic memory search.
GET	/patients/{patient_id}/conversations	Patient/Caregiver	List permitted conversation metadata.
DELETE	/conversations/{conversation_id}	Patient/Caregiver	Delete/retire conversation according to retention policy.

10. Personal AI Assistant API

Method	Endpoint	Auth	Purpose
POST	/assistant/sessions	Patient	Start assistant session.

POST	/assistant/query	Patient	Ask typed/voice-derived question.
GET	/assistant/sessions/{session_id}	Patient	Retrieve session metadata.
POST	/assistant/sessions/{session_id}/end	Patient	End session.

11. Assistant Query Contract

Example request:

```
{"patient_id":"...", "query":"Where did I keep my keys?", "mode":"text",
"context_person_id":null}
```

Example response:

```
{"success":true,"data":{"answer":"Your keys were last detected near the study
table.","sources":[{"type":"object_observation","id":"...","observed_at":"..."}],"confidence":"me
dium"}}
```

Important behavior: the assistant should use deterministic APIs for schedule/medication facts and RAG/vector retrieval for conversational memory. It should not invent missing information.

12. Geofencing and Location APIs

Method	Endpoint	Auth	Purpose
GET	/patients/{patient_id}/geofences	Caregiver	List safe zones.
POST	/patients/{patient_id}/geofences	Caregiver	Create safe zone.
PATCH	/geofences/{geofence_id}	Caregiver	Update safe zone.
DELETE	/geofences/{geofence_id}	Caregiver	Deactivate safe zone.
POST	/patients/{patient_id}/locations	Patient/System	Submit location event.
GET	/patients/{patient_id}/locations/latest	Caregiver	Get latest location.
GET	/patients/{patient_id}/alerts	Caregiver/Patient	List alerts.
POST	/alerts/{alert_id}/acknowledge	Caregiver	Acknowledge alert.

13. Well-being APIs

Method	Endpoint	Auth	Purpose
GET	/wellbeing/content	Patient	List meditation/relaxation content.
GET	/wellbeing/content/{content_id}	Patient	Get content metadata.
POST	/patients/{patient_id}/wellbeing/sessions	Patient	Record a session start.
POST	/wellbeing/sessions/{session_id}/end	Patient	Record session completion.

14. Activity and Audit APIs

Method	Endpoint	Auth	Purpose
GET	/patients/{patient_id}/activity	Caregiver/Patient	View permitted activity history.
GET	/patients/{patient_id}/audit	Authorized admin/caregiver	View permitted security/configuration events.

15. Request Validation

API Area	Required Validation
Auth	Credential presence; account active; rate limiting where deployed.
Medication	Name required; schedule valid; patient authorization.
Appointment	Title/start time required; patient authorization.

Schedule	Title/time/recurrence required.
Person	Name required; patient authorization.
Face	Supported image/input; person registration authorization.
Object	Supported media; object confidence threshold; patient context.
Memory	Patient context; source; content/vector reference.
Geofence	Valid coordinates; positive radius; caregiver authorization.
Location	Valid coordinates and timestamp; authenticated patient/device.
Assistant	Non-empty query; authenticated patient; context authorization.

16. HTTP Status Codes

Code	Meaning
200	Successful read/update/action.
201	Resource created.
204	Successful deletion/deactivation with no body.
400	Validation error.
401	Unauthenticated/invalid token.
403	Authenticated but not authorized.
404	Resource not found or not visible to caller.
409	Conflict/duplicate.
422	Semantically invalid request if used by framework.
429	Rate limit exceeded.
500	Unexpected server error.
503	Dependent AI/notification service unavailable.

17. Security Requirements for APIs

- Every protected endpoint must perform authentication.
- Every patient-specific endpoint must enforce patient/caregiver authorization.
- Do not trust patient_id supplied by the client without authorization checks.
- Do not expose raw face embeddings to ordinary clients.
- Do not expose internal vector-store identifiers unless needed.
- Validate uploaded media type and size.
- Apply rate limits to authentication, face recognition and assistant endpoints where practical.
- Log security-sensitive actions without logging unnecessary sensitive content.

18. API Dependency Flow

1. Mobile app authenticates → receives access token.
2. Mobile app requests schedule/medication → backend queries relational database.
3. Camera frame → object/face service → backend stores result.
4. Conversation → STT → memory ingestion → embedding/vector store.
5. Assistant query → intent/context routing → structured DB and/or vector search → LLM → response.
6. GPS event → geofence evaluator → alert record → notification.

19. API Implementation Order

7. Auth + users + RBAC
8. Patient/caregiver authorization
9. Schedule + appointments
10. Medication + reminders
11. People + face registration
12. Objects + observations

13. Conversations + memory ingestion
14. Memory search + assistant
15. Geofences + locations + alerts
16. Well-being + activity logging

20. API Completion Checklist

- OpenAPI/Swagger specification generated from final routes.
- Authentication and authorization tested on every protected endpoint.
- Request/response schemas documented.
- Error responses standardized.
- Pagination added to list endpoints where required.
- Rate limiting/security controls reviewed.
- Integration tests mapped to requirement IDs.
- Frontend API client generated/implemented.